From Simulation to Experimental Analysis

Multi-Trial EMG & Kinematic Data with Pandas and Seaborn

Week 2 Lecture — Machine Learning for Neuroscience

Building on Week 1: EMG-Driven Reaching Movements

Lecture Outline

1. From Arrays to DataFrames: Structuring Experimental Data
2. The Tidy Data Paradigm for Motor Control Experiments
3. GroupBy Operations: Trial-Level to Condition-Level Analysis
4. Statistical Visualization with Seaborn
5. Trial-Averaged Kinematics and Confidence Intervals
6. Feature Correlations and Exploratory Data Analysis
7. Comparing Populations: Healthy vs. Impaired Subjects
8. Building Publication-Quality Figures

1. From Arrays to DataFrames

In Week 1, we generated EMG signals and kinematic trajectories as **NumPy arrays**. This is efficient for computation, but real experiments produce data that must be organized by subject, trial, condition, and time. **Pandas DataFrames** provide the structure needed to manage this complexity.

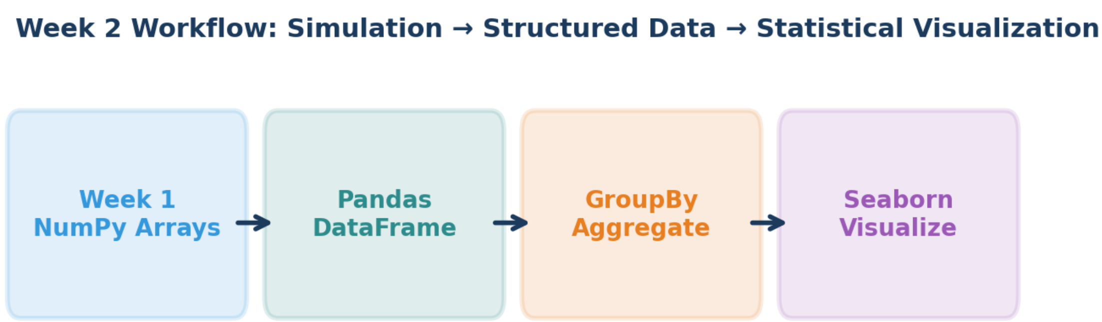

Figure 1: Week 2 workflow — NumPy simulation outputs are structured into DataFrames for statistical analysis.

The key insight is that each row in a DataFrame represents **one observation**. For our reaching experiment, we create two complementary tables:

- **Summary table:** One row per trial with extracted metrics (endpoint error, movement time, peak velocity, etc.)
- **Time-series table:** One row per time point per trial with raw signals (angle, velocity, EMG)

Tidy DataFrame Structure: One Row Per Trial

subject_id	condition	trial	endpoint_erro	movement_time_n	peak_velocity
S01	slow	1	6.9	665.0	346.62
S01	slow	2	3.65	737.0	401.49
S01	slow	3	10.04	607.0	325.7
S01	slow	4	13.43	588.0	306.23
S01	slow	5	6.94	675.0	341.66
S01	slow	6	14.75	712.0	317.84
S01	slow	7	8.89	737.0	412.02
S01	slow	8	6.32	578.0	360.0

Figure 2: The tidy DataFrame structure. Each row is one trial; columns are variables.

> **➔ Key Concept: Tidy data (Wickham, 2014) means each variable is a column, each observation is a row, and each type of observational unit forms a table. This structure unlocks the full power of Pandas groupby and Seaborn plotting.**

2. The Virtual Experiment

We simulate a realistic motor control experiment by running the Week 1 model as a batch:

Parameter	Value	Purpose
Subjects	10 healthy + 5 impaired	Between-subject comparisons
Conditions	Slow / Normal / Fast	Speed-accuracy tradeoff
Trials per condition	30	Within-subject variability
Total trials (healthy)	900	Statistical power
Trial-to-trial noise	8% amplitude, 15 ms timing	Realistic motor variability
Impaired group	40 ms ANT timing noise	Simulates cerebellar dysfunction

The **trial-to-trial variability** is critical for realism. Even well-practiced movements show ~5-10% variability in peak force and ~10-20 ms variability in burst timing. This noise is signal-dependent (Harris & Wolpert, 1998) and is the fundamental source of the speed-accuracy tradeoff.

→ Key Concept: Every motor neuroscience experiment must deal with inter-trial variability. Pandas groupby operations let us compute trial-averaged metrics and confidence intervals, separating the signal (condition effects) from the noise (trial-to-trial fluctuations).

3. GroupBy: The Core of Experimental Analysis

The **split-apply-combine** pattern is the workhorse of experimental data analysis:

- **Split:** Partition data by condition (or subject, or both)
- **Apply:** Compute statistics (mean, std, SEM) within each group
- **Combine:** Reassemble into a summary table

Example Pandas code:

```
df.groupby('condition')['peak_velocity'].agg(['mean', 'std', 'count'])
df.groupby(['subject_id', 'condition']).agg({
'peak_velocity': 'mean', 'endpoint_error': ['mean','std']})
```

The **pivot table** is a powerful extension that creates cross-tabulations:

```
pd.pivot_table(df, values='peak_velocity', index='subject_id',
columns='condition', aggfunc='mean')
```

This creates a subjects × conditions matrix — exactly the format needed for repeated-measures statistical tests or ML feature matrices in later weeks.

4. Statistical Visualization with Seaborn

Seaborn is built on top of Matplotlib but provides **statistical plotting** functions that automatically compute and display uncertainty. Key plot types for motor control data:

Plot Type	Seaborn Function	Best For
Violin + strip	violinplot + stripplot	Distribution shape + individual trials
Box plot	boxplot	Median, quartiles, outliers
Line with CI	lineplot	Time-series with confidence bands
Pair plot	pairplot	All pairwise feature relationships
Heatmap	heatmap	Correlation matrices
Facet grid	FacetGrid	Same plot across subjects/conditions

Figure 3: Violin plots with strip plots show both the distribution shape and individual trial data points.

5. Trial-Averaged Kinematics

A hallmark of motor control publications is the **trial-averaged velocity profile** — the mean trajectory with shaded confidence bands. Seaborn's lineplot computes this automatically:

```
sns.lineplot(data=df_ts, x='time_ms', y='omega_deg_s', hue='condition')
```

By default, Seaborn computes the mean and 95% confidence interval across all observations at each time point. This reveals the **stereotypical velocity profile** buried within noisy individual trials.

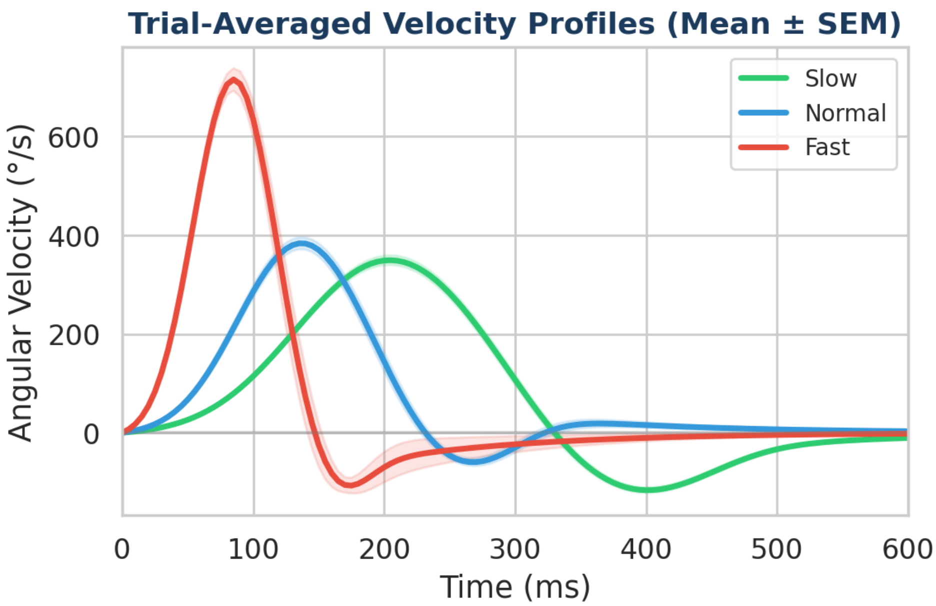

Figure 4: Trial-averaged velocity profiles. The bell-shaped pattern is preserved across conditions, but faster movements show higher peaks and shorter durations.

➔ Key Concept: The standard error of the mean (SEM = SD / √N) decreases with more trials, reflecting increased confidence in the population mean. This is why motor control experiments typically collect 20-50 trials per condition.

6. Feature Correlations and EDA

Exploratory data analysis (EDA) reveals the relationships between features before formal modeling. Two essential tools:

Pair Plots

Pair plots show every pairwise combination of features, colored by condition. This immediately reveals which features covary and which are independent — crucial information for feature selection in ML models (Weeks 4-5).

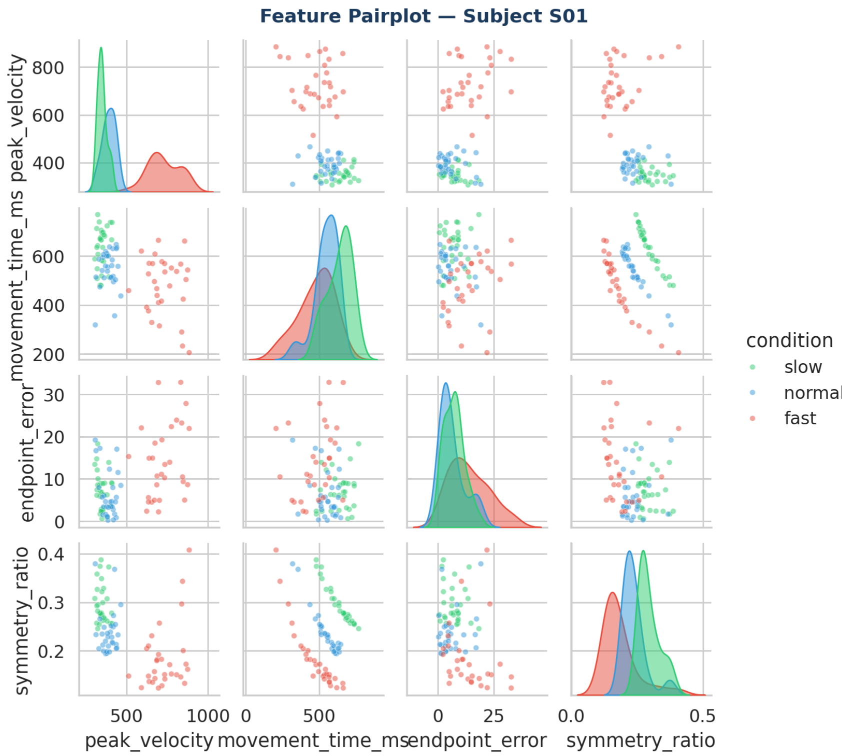

Figure 5: Pair plot reveals the multivariate structure of movement features.

Correlation Heatmaps

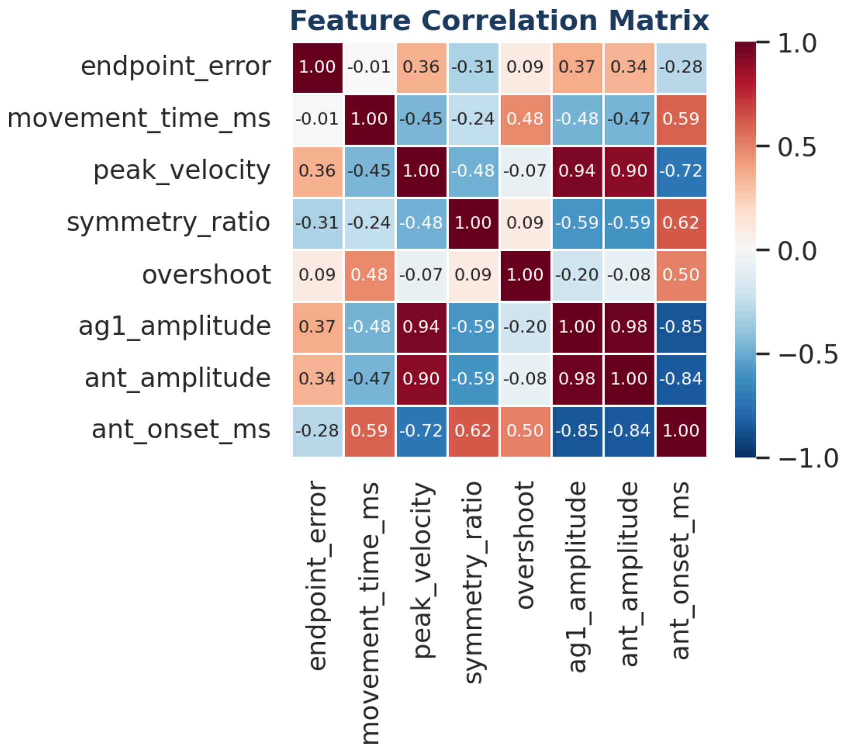

Figure 6: Correlation matrix. Strong correlations (red/blue) indicate redundant features; near-zero values indicate independent information.

Note how peak velocity and AG1 amplitude are strongly correlated (— this makes biomechanical sense, since more agonist force produces faster movement). Identifying these redundancies now will inform **feature selection** in Week 4.

7. Comparing Populations

A major goal of motor neuroscience is comparing movement patterns between clinical populations. We simulate this by creating **impaired subjects** with increased antagonist timing variability (SD = 40 ms vs. 15 ms) — modeling cerebellar ataxia.

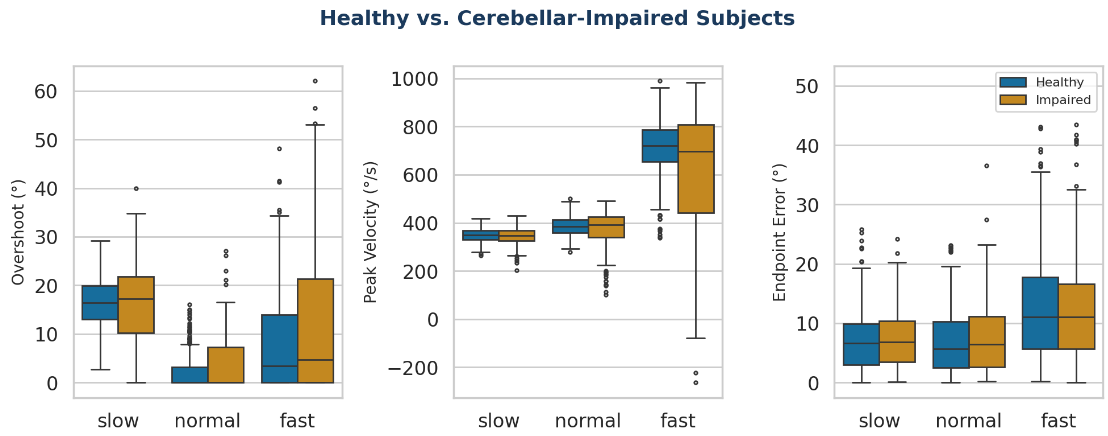

Figure 7: Healthy vs. impaired comparison. Impaired subjects show greater overshoot and more variable kinematics.

The impaired group shows systematically larger overshoot, higher variability in peak velocity, and increased endpoint error — all consistent with the **dysmetria** pattern described in Week 1. This comparison foreshadows **classification tasks** in later weeks: can an ML model learn to distinguish healthy from impaired subjects based on kinematic features?

> **→ Key Concept: Visualizing group differences with Seaborn (box plots, violin plots, overlapping distributions) is the first step in any classification pipeline. If groups are visually separable, an ML model will likely succeed. If they overlap heavily, more sophisticated features or models may be needed.**

8. Building Publication-Quality Figures

Motor control papers typically present results as multi-panel figures. The key principles:

- **Consistent color coding** across panels (same color = same condition everywhere)
- **Shared axes** where appropriate (e.g., shared time axis for EMG and kinematics)
- **Individual trials + group summary** (show raw data, not just means)
- **Error bars that match the inference**: SEM for population means, SD for individual variability
- **Panel labels** (A, B, C, D) with informative subtitles

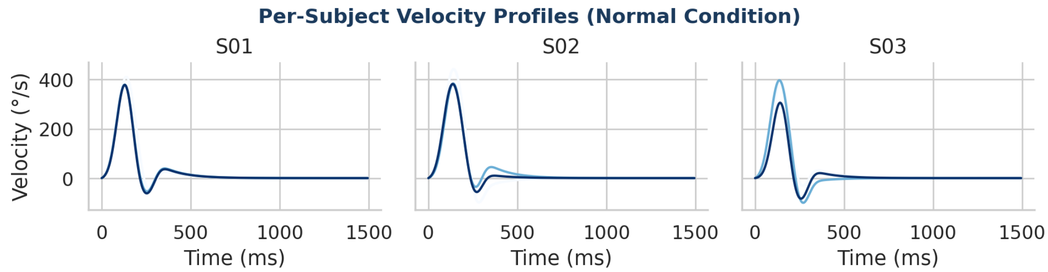

Figure 8: Seaborn FacetGrid showing per-subject velocity profiles. This reveals individual differences that are hidden in group averages.

Essential Seaborn Setup

```
sns.set_theme(style='whitegrid', palette='colorblind', font_scale=1.1)
```

```
plt.savefig('figure.png', dpi=300, bbox_inches='tight')
```

> **→ Today's Lab: You will build this entire analysis pipeline — from generating the multi-trial dataset to producing publication-quality figures comparing conditions and subject groups. See the lab notebook for hands-on exercises.**

Key References

Harris, C.M. & Wolpert, D.M. (1998). Signal-dependent noise determines motor planning. *Nature*, 394, 780-784.

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McKinney, W. (2010). Data structures for statistical computing in Python. *Proc. 9th Python in Science Conf.*, 51-56.

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